

CSC1003 Midterm Review

Introduction to Computer Science and Java Programming

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Compile

javac/java

```
1 javac [options] [source files]
2 java [options] classname [args]
```

.java $\xrightarrow{\text{Java Compiler (javac)}}$.class JVM executable

args/Scanner

- args: Static input - provided at program startup
- Scanner: Dynamic input - collected during program execution

Quiz

```
1 public class quiz {
2     public static void main(String[] args) {
3         String output = "";
4         for (int i = 0; i < args.length; i++) {
5             if (args[i].length() > 3) {
6                 output += args[i].charAt(3);
7             }
8             else {
9                 output += args[i];
10            }
11        }
12        System.out.println(output);
13    }
14 }
```

```
1 java quiz apple banana cat dog CSC1003
```

Primitive Type

- byte $[-2^7, 2^7 - 1]$
- short $[-2^{15}, 2^{15} - 1]$
- int $[-2^{31}, 2^{31} - 1]$
- long $[-2^{63}, 2^{63} - 1]$
- float 32-bit floating point
- double 64-bit floating point
- char 16-bit Unicode character
- boolean true/false values

Data Types

Reference Types

- Classes (e.g., String) *String不是一个数值类型，是一种语法*
- Interfaces
- Arrays

```
1 String s = "CSC1003";
```

Reference Types

- Classes (e.g., String)
- Interfaces
- Arrays

```
1 String s = "CSC1003";
```

```
1 String s = new String("abc");  
2 String s1 = String.valueOf(100);  
3 String s2 = Double.toString(3.14);
```

Quiz

```
1 public class quiz {  
2     public static void main(String[] args) {  
3         String a = 1 + 2 + "3";  
4         String b = 1 + "2" + 3;    从左往右算  
5         String c = "1" + 2 + 3;  
6         System.out.println (a + b + c);  
7     }  
8 }
```

Formatted Output

```
System.out.printf ();
```

Specifier	Purpose	Example
%s	String	System.out.printf("%s",str);
%d	Integer	System.out.printf("%d",val);
%c	Character	System.out.printf("%c",ch);
%f	Float (6)	System.out.printf("%f",val);
%.nf	n Decimal	System.out.printf("%.1f",1.23);

Note: Rounding Behavior

\n代表回车

- %.nf automatically rounds to n decimal places
- Example: %.1f rounds 1.25 → 1.3, 1.24 → 1.2
- Uses standard half-up rounding (0.5 rounds up)

保留一位小数, 自动四舍五入

if Statement

Base Syntax

```
1  if (condition) {  
2      //if the condition is true  
3  }  
4  else {  
5      //otherwise  
6  }
```

Ternary Operator

```
1  condition ? exprIfTrue : exprIfFalse
```

Quiz

```
1 public class quiz {  
2     public static void main(String[] args) {  
3         int a = 15, b = 25, c = 10;  
4         int val = (a > b) ?  
5             ((a > c) ? a : c) :  
6             ((b > c) ? b : c);  
7         System.out.println (val);  
8     }  
9 }
```

while Statement

Base Syntax

```
1 while (condition){  
2     //repeat if condition is true  
3 }
```

```
1 do{  
2     //execute at least once  
3     //repeat if condition is true  
4 } while (condition);
```

先执行，再判断

Note: The semicolon ; after the while condition is mandatory!

for Statement

Base Syntax

```
1 for (initialization; condition; update){  
2     //repeat until condition is false  
3 }
```

Quiz

```
1 public class quiz {
2     public static void main(String[] args) {
3         long n = Long.parseLong(args[0]);
4         for (int i = 2; i <= n / i; i++) {
5             while (n % i == 0) {
6                 System.out.print(i + ",");
7                 n = n / i;
8             }
9         }
10        if (n > 1) System.out.println(n);
11        else System.out.println();
12    }
13 }
```

建议：每次写完都输出换行

```
1 java quiz 228
```

Fundamental Operations

Operation	Array Code
Default initialization	<code>arr = new int[1000];</code>
Declare and initialize	<code>int[] arr = new int[1000];</code>
Literal initialization	<code>int[] arr = {1,2,3};</code> <code>int[] arr = new int[] {1,2,3};</code>

`int[] arr = new int[3] = {1,2,3};`
Array constants can only be used in initializers!

后面看出元素个数, 前面不用声名长度

Comparison with Strings

Array	String
<code>arr.length</code>	<code>str.length()</code>
<code>element = arr[index]</code>	<code>char = str.charAt(index)</code>
Mutable	Immutable

```
1  int[] a = {1, 2, 3};    此处a和b是关联的，一个动，会同时改变两个
2  int[] b = a;
3  int[] c = a.clone ();  将a中元素都克隆了一份
4  System.out.println (a == b);  而c只是克隆了值并成为一个新的数组，与ab无关
5  System.out.println (a == c);
6  b[0] = 99; c[1] = 99;
7  System.out.println (a[0]);
8  System.out.println (a[1]);
```

Quiz

```
1  for (int i = 0; i < n / 2; i++) {  
2      int x = n - i - 1;  
3      for (int j = i; j < x; j++) {  
4          int y = n - j - 1;  
5          int t = a[i][j];  
6          a[i][j] = a[y][i];  
7          a[y][i] = a[x][y];  
8          a[x][y] = a[j][x];  
9          a[j][x] = t;  
10     }  
11 }
```

可以通过画表格模拟

本质是旋转复制

Method Overloading

- Same method name with different parameters is allowed
- Parameters must differ in type, number, or order
- Return type alone cannot differentiate overloaded methods
- Compiler selects appropriate method based on arguments

```
1 public static void 无返回值print (int number) { }  
2 public static void print (String text) { }  
3 public static int count (int a) { }  
4 public static int count (int a,int b) { }  
5 public static int cmp (int a,int b) { }  
6 public static int cmp (int b,int a) { }  
7 public static int calc (int a,int b) { } 看括号里的是否相同  
8 public static double calc (int a,int b) { }
```

Functions

continue/break/return

Statement	Effect
break	Exits the current loop or switch statement immediately
continue	Skips the current iteration and moves to the next one
return	Exits the method and optionally returns a value

在void中相当于终止

```
1 public static int calc (int n) {  
2     if (n <= 0) return -1;  
3     int sum = 0;  
4     for (int i = 1; i <= n; i++) {  
5         if (i % 2 == 1) continue; 跳过单次循环  
6         if (i * i > n) break;  
7         sum += i;                一层一层跳, 只跳出一层  
8     }  
9     return sum;  
10 }
```

Practice

```
1 public class Q1 {
2     public static void main(String[] args) {
3         int s = 0;
4         for (int i = 0; i < args.length; i++)
5             s += Integer.parseInt(args[i]);
6         int a = s / args.length;
7         double b = s / args.length;
8         System.out.println(a);
9         System.out.printf("%.1f\n", b);
10    }
11 }
```

Questions

- What is the command to compile the program?
- What is the output when running: `java Q1 1 2 3 4`?

Practice

```
1 public class Q2 {
2     public static void main(String[] args) {
3         int[] array = new int[args.length];
4         int start = 0, end = args.length - 1;
5         while (start < end) {
6             int temp = Integer.parseInt(args[start]);
7             array[start] = Integer.parseInt(args[end]);
8             array[end] = temp;
9             start++; end--;
10        }
11        for (int i = 0; i < array.length; i++) {
12            System.out.print(array[i]);
13            if (i != array.length - 1) {
14                System.out.print(", ");
15            }
16        }
17    }
18 }
```

Questions

- What is the output when running: `java Q2 1 2 3 4 5 > result.txt`?
- If the program should reverse the input numbers, point out the mistake and correct it.

Questions

- What is the output when running: `java Q2 1 2 3 4 5 > result.txt`?
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Reverse the array

```
1 for (int i = 0; i < n / 2; ++i) {  
2     int temp = arr[i];  
3     arr[i] = arr[n - i - 1];  
4     arr[n - i - 1] = temp;  
5 }
```

对称位置从而进行翻转

Practice

```
1 public class Q3 {
2     public static void main(String args[]) {
3         int n = Integer.parseInt(args[0]);
4         int sum = 0, temp = n;
5         while (n > 0) {
6             sum = sum * 10 + n % 10;  取个位
7             n /= 10;  向下取整
8         }
9         if (temp == sum) System.out.println("true");
10        else System.out.println("false");
11    }
12 }
```

Questions

- What is the output when running: `java Q3 121212?`
- What is the output when running: `java Q3 121121?`

Palindrome

```
1 public class Q3 {  
2     public static void main(String args[]) {  
3         int n = Integer.parseInt(args[0]);  
4         int sum = 0;  
5         while (n > sum) {  
6             sum = sum * 10 + n % 10;  
7             n /= 10;  
8         }  
9         if (____1____) System.out.println("true");  
10        else System.out.println("false");  
11    }  
12 }
```

Questions

- Complete the missing code at line 9.
- Any errors in the code? ($n \in \mathbb{N}$)

Practice

```
1 public class Q4 {  
2     public static void main(String[] args) {  
3         int x = Integer.parseInt(args[0]);  
4         int y = Integer.parseInt(args[1]);  
5         int g = 1;  
6         for (int i = 1; i <= x && i <= y; i++) {  
7             if ((x % i) == 0 && (y % i) == 0)  
8                 g = i;  
9         }  
10        System.out.println(g);  
11    }  
12 }
```

Questions

- What is the output when running: `java Q4 24 84`?

```
1 public class Q4 {  
2     public static void main(String[] args) {  
3         int x = Integer.parseInt(args[0]);  
4         int y = Integer.parseInt(args[1]);  
5         int l = x * y;  
6         for (int i = x * y; i >= x && i >= y; i--) {  
7             if (i % x == 0 && i % y == 0)  
8                 l = i;  
9         }  
10        System.out.println(l);  
11    }  
12 }
```

最小公倍数

Questions

- What is the output when running: `java Q4 24 84`?

Practice

```
1 public class Q6 {  
2     public static void main(String[] args) {  
3         int[] a = new int[args.length];  
4         int cnt = 0;  
5         for (int i = 0; i < args.length; i++)  
6             a[i] = Integer.parseInt(args[i]);  
7         for (int i = 0; i < a.length; i++)  
8             for (int j = i + 1; j < a.length; j++)  
9                 if (a[i] > a[j]) {  
10                     int temp = a[i]; a[i] = a[j]; a[j] = temp;  
11                     ++cnt;  
12                 }  
13         for (int i = 0; i < a.length; i++)  
14             System.out.print((a[i] + cnt) + " ");  
15     }  
16 }
```

Question

What is the output when running: `java Q6 3 8 9 5 2 4 1?`

Sort

```
1  for (int i = 0; i < a.length-1; i++) {  
2      int minIndex = i;  
3      for (int j = i+1; j < a.length; j++)  
4          if (a[j] < a[minIndex]) minIndex = j;  
5      int temp = a[minIndex];  
6      a[minIndex] = a[i];  
7      a[i] = temp;  
8  }
```

```
1  for (int i = 1; i < a.length; i++) {  
2      int key = a[i], j = i - 1;  
3      while (j >= 0 && a[j] > key) {  
4          a[j+1] = a[j];  
5          j--;  
6      }  
7      a[j+1] = key;  
8  }
```

Practice

Consider a program that takes a positive integer n as input and prints the binary representation of n .

```
1 public class Q7 {  
2     public static void main(String[] args) {  
3         int n = Integer.parseInt(args[0]), power = 1;  
4         while (power < n/2) power *= 2;  
5         while (power > 0) {  
6             if (n < power) System.out.print(0);  
7             else {System.out.print(1); n -= power;}  
8             power /= 2;  
9         }  
10    }  
11 }
```

Question

The program has one mistake. Point out the line number and correct it.

Practice

```
1 public class Q8 {  
2     public static void main(String[] args) {  
3         int n = 1000;  
4         int[] array = new int[n];  
5         for (int i = 0; i < n; i++)  
6             array[i] = (int) Math.round(Math.random() * 255); //[0,255] in random  
7         int[] counts = new int[255];  
8         for (int i = 0; i < n; i++) _____  
9         for (int i = 0; i < counts.length; i++) //Outputs the frequencies.  
10            System.out.println(i + ":" + counts[i]);  
11    }  
12 }
```

Question

- Point out the line number of the mistake and correct it.
- Complete the missing code at line 8 so the program correctly counts frequencies.

Practice

```
1 public class Q9 {  
2     public static int f(int[] a, int n) {  
3         int i1 = 0, i2 = a.length, c = 0;  
4         while (true) {  
5             c++;  
6             int i = (i1 + i2) / 2;  
7             if (a[i] == n) return c;  
8             if (c > a.length) return -1;  
9             if (a[i] < n) i1 = i;  
10            if (a[i] > n) i2 = i;  
11        }  
12    }  
13    public static void main(String[] args) {  
14        int n = Integer.parseInt(args[0]);  
15        int[] array = new int[100];  
16        for (int i = 0; i < array.length; i++) array[i] = i;  
17        System.out.println("c=" + f(array, n));  
18    }  
19 }
```

2分过程

Question

- What is the output when running: `java Q9 31`?
- What is the output when running: `java Q9 128`?
- For input integers between 0 and 99, what is the largest output value of `c`?

Binary Search

```
1 public static void main(String[] args) {  
2     int n = Integer.parseInt(args[0]);  
3     int[] array = new int[n];  
4     for (int i = 0; i < array.length; i++) array[i] = i;  
5     int cnt = 0, _max = (int) Math.ceil(Math.log(n) / Math.log(2));  
6     for (int i = 0; i < n; i++)  
7         if (f(array, i) == _max) ++cnt;  
8     System.out.println (cnt);  
9 }
```

Binary Search

```
1 public static void main(String[] args) {  
2     int n = Integer.parseInt(args[0]);  
3     int[] array = new int[n];  
4     for (int i = 0; i < array.length; i++) array[i] = i;  
5     int cnt = 0, _max = (int) Math.ceil(Math.log(n) / Math.log(2));  
6     for (int i = 0; i < n; i++)  
7         if (f(array, i) == _max) ++cnt;  
8     System.out.println (cnt);  
9 }
```

Answer

Let $k = \lfloor \log_2(n) \rfloor$.

The number of midpoints discovered in at most k iterations is
 $2^0 + 2^1 + \dots + 2^{k-1} = 2^k - 1$.

Thus the remaining indices (found first at iteration $k + 1$) are
 $n - (2^k - 1) = n - 2^{\lfloor \log_2(n) \rfloor} + 1$.

- Ready to become a coding **MASTER**?
- Don't miss the school competition next semester!



Q & A

Thanks for listening!

Good Luck on Your Midterm Exams!